PROBLEM SOLVING

DAY 6

Basic Decision & Sequence

1. Even or Odd

Design a flowchart to check whether a number is even or odd.

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2. Largest of Two Numbers

Input two numbers, output the larger one.

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3. Positive / Negative / Zero

Check whether a number is positive, negative, or zero.

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4. Sum of First N Numbers

Compute sum from 1 to n.

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5. Multiplication Table

Print multiplication table of a number up to 10.

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6. Simple Calculator

Input two numbers and an operator (+, -, *, /) → output result.

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7. Swap Two Numbers

Swap two numbers (with and without temp variable).

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8. Count Digits

Count number of digits in a number.

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9. Reverse a Number

Reverse digits of a number.

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10. Check Divisibility

Check if a number is divisible by both 3 and 5.

Loops + Nested Logic

11. Factorial

Calculate factorial of a number using loop.

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12. Fibonacci Series

Print first n Fibonacci numbers.

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13. Prime Check

Check whether a number is prime.

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14. Sum of Digits Until Single Digit

Repeat sum of digits until one digit remains.

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15. Palindrome Number

Check if a number is palindrome.

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16. Armstrong Number

Check if a number is Armstrong.

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17. Count Even and Odd Digits

Count even and odd digits in a number.

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18. Find Largest in Array

Input n elements → find maximum.

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19. Linear Search

Search for element x in array.

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20. Pattern Printing

Print pattern:

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Optimization Thinking

21. GCD (Euclidean Algorithm)

Design optimized flowchart using repeated modulo.

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22. Prefix Sum Concept

Given array, answer sum queries efficiently.

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23. Count Frequency of Elements

Use logic to count occurrences.

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24. Find Second Largest

Without sorting.

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25. Remove Duplicates

From array (logic-based, not built-in).

Multi-Step Logic & Efficiency

26. Two Sum Problem

Find two numbers that add to target.

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27. Maximum Subarray Sum

Design flowchart for Kadane’s Algorithm.

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28. Binary Search

Search element in sorted array.

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29. Merge Two Sorted Arrays

Into one sorted array.

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30. Subarray with Given Sum

Efficient approach (sliding window idea).
